CAP 3330. Fall 2026

Assignment 3 (100 points)

Instructions

Create an R notebook and include all the **code** and **interpretations/explanations** needed to answer the **two questions** below.

Question 1 (60 points)

A researcher is interested in studying three types of fertilization methods (100 lb., 150 lb., and 200 lb.) and two levels of irrigation (A and B) on biomass yield. The possible treatment combinations were randomly assigned to 30 plots of land, where each treatment was assigned the same number of plots. You can find the data you need for this exercise in the file **Biomass.csv** (Canvas/ Module 3/ Resources).

Note: You **MUST** read the Biomass.csv file into R using the following way:

- Open a code chunk in your notebook, insert the following code, and run the code chunk:

```
biomass_df = read.csv ("Biomass.csv",colClasses = c("numeric","factor","factor"))
```

Once you do that, a data frame called "biomass_df" will be available in R with the data you need to answer this question.

- *If you attempt to read the file in a different way* (for example, the way we do it in class using the Import option from the Environment tab), you ***will likely*** read the file incorrectly and your results will be wrong.

a) How many replications were taken in this study? **Show** how you got the answer.

b) Identify:

- The outcome variable
- The factor (or factors)
- The levels of the factor (or levels of the factors)

c) Set up all the hypotheses that should be set up to run this ANOVA.

d) Analyze the results and discuss which effects (among all possible effects) are statistically significant. **Justify** your answer.

e) Construct the interaction plot and **explain** why it clearly shows the presence of interaction in this problem.

Note: Stating that *"the interaction is evident because the lines are not parallel"* (or a similar statement based on observing that the lines are not parallel) does NOT count as a valid explanation.

Question 2 (40 points)

A study is done to determine if there is a difference in the average strength of a filament fiber produced by three machines. Researchers are also interested in studying the possible effect of changing the filament diameter on strength. Researchers decided to do the analysis using an alpha of **0.10.**

The data for this exercise can be found in the file **Filament.csv** (Canvas/ Module 3/ Resources). You **MUST** read the Filament.csv file into R using the following way:

Open a code chunk in your notebook, insert the following code, and run the code chunk:

```
Filament_df = read.csv ("Filament.csv",colClasses = c("numeric","factor","factor"))
```

a) Identify:

- The outcome variable
- The factor (or factors)
- The levels of the factor (or levels of the factors)

b) Analyze the results and discuss which effects (among all possible effects) are statistically significant. **Justify** your answer.

c) Conduct a post-hoc test for the machine factor using the Tukey method (only for the machine factor !). Discuss which machines lead to statistically different average strength. **Explain and show your work.**